

● EASY

● ALGEBRA

● ANSWER KEY

# Algebra

# 10

10 answers · Rationale-rich · Teach from doc

**Q1****C**

**C.**  $31 \leq t \leq 67$

Choice C is correct. It's given that for a 3-week period in a town in Illinois, the lowest recorded temperature was 31°F and the highest recorded temperature was 67°F. It follows that the inequality  $31 \leq t \leq 67$  is true for any recorded temperature  $t$ , in °F, in this town for this 3-week period.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

**Q2****B**

**B.**  $y = -3x + 3$

Choice B is correct. Any line in the  $xy$ -plane can be defined by an equation in the form  $y = mx + b$ , where  $m$  is the slope of the line and  $b$  is the  $y$ -coordinate of the  $y$ -intercept of the line.

From the graph, the  $y$ -intercept of the line is  $(0, 3)$ . Therefore,  $b = 3$ . The slope of the line is the change in the value of  $y$  divided by the change in the value of  $x$  for any two points on the

line. The line shown passes through  $(0, 3)$  and  $(1, 0)$ , so  $m = \frac{3 - 0}{0 - 1}$ , or  $m = -3$ . Therefore,

the equation of the line is  $y = -3x + 3$ .

Choices A and C are incorrect because the equations given in these choices represent a line with a positive slope. However, the line shown has a negative slope. Choice D is incorrect because

the equation given in this choice represents a line with slope of  $-\frac{1}{3}$ . However, the line shown has a slope of  $-3$ .

**Q3****27**

The correct answer is 27. Multiplying both sides of the given equation by 3 yields  $36 + x = 39$ , or  $18 + 3x = 27$ . Therefore, the value of  $18 + 3x$  is 27.

**Q4****C**

C.  $c = 23t + 35$

Choice C is correct. It's given that this service contract requires a monthly cost of \$ 23. A monthly cost of \$ 23 for  $t$  months results in a cost of \$  $23t$ . It's also given that this service contract requires a onetime activation cost of \$ 35. Adding the onetime activation cost to the monthly cost of the service contract for  $t$  months yields the total cost  $c$ , in dollars, of this service contract for  $t$  months. Therefore, this situation can be represented by the equation  $c = 23t + 35$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q5****A**

A. 4, 28

Choice A is correct. The second equation in the given system is  $y = 28$ . Substituting 28 for  $y$  in the first equation in the given system yields  $28 = 12x - 20$ . Adding 20 to both sides of this equation yields  $48 = 12x$ . Dividing both sides of this equation by 12 yields  $4 = x$ . Therefore, the solution  $x, y$  to the given system of equations is 4, 28.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the solution  $y, x$ , not  $x, y$ , to the given system of equations.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q6****B**

B.  $35r + 40j < 220$

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Choice B is correct. It's given that Rhett drove at an average speed of 35 miles per hour and that he drove for  $r$  hours. Multiplying 35 miles per hour by  $r$  hours yields  $35r$  miles, or the distance that Rhett drove. It's also given that Jessica drove at an average speed of 40 miles per hour and that she drove for  $j$  hours. Multiplying 40 miles per hour by  $j$  hours yields  $40j$  miles, or the distance that Jessica drove. The total distance, in miles, that Rhett and Jessica drove can be represented by the expression  $35r + 40j$ . It's given that the total distance they drove was under 220 miles. Therefore, the inequality  $35r + 40j < 220$  represents this situation.

Choice A is incorrect. This inequality represents a situation in which the total distance Rhett and Jessica drove was over, rather than under, 220 miles.

Choice C is incorrect. This inequality represents a situation in which Rhett drove at an average speed of 40, rather than 35, miles per hour, Jessica drove at an average speed of 35, rather than 40, miles per hour, and the total distance they drove was over, rather than under, 220 miles.

Choice D is incorrect. This inequality represents a situation in which Rhett drove at an average speed of 40, rather than 35, miles per hour, and Jessica drove at an average speed of 35, rather than 40, miles per hour.

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**Q7****9**

The correct answer is 9. It's given that the  $y$ -intercept of the graph of  $12x + 2y = 18$  in the  $xy$ -plane is  $0, y$ . Substituting 0 for  $x$  in the equation  $12x + 2y = 18$  yields  $12(0) + 2y = 18$ , or  $2y = 18$ . Dividing both sides of this equation by 2 yields  $y = 9$ . Therefore, the value of  $y$  is 9.

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**Q8****10**

The correct answer is 10. It's given that the cost for the entire purchase was \$ 27 after a coupon was used for \$ 63 off the entire purchase. Adding the amount of the coupon to the purchase price yields  $27 + 63 = 90$ . Thus, the cost for the entire purchase before using the coupon was \$ 90. It's given that Nasir bought 9 storage bins. The original price for 1 storage bin can be found by dividing the total cost by 9. Therefore, the original price, in dollars, for 1 storage bin is  $\frac{90}{9}$ , or 10.

**Q9****B****B. 13**

Choice B is correct. It's given that the function  $f$  is defined by  $fx = 3x - 8$ . The value of  $f7$  is the value of  $fx$  when  $x = 7$ . Substituting 7 for  $x$  in the given equation yields  $f7 = 37 - 8$ , which is equivalent to  $f7 = 21 - 8$ , or  $f7 = 13$ .

Choice A is incorrect. This is the value of  $f7$  when  $fx = 3x + 8$ , rather than  $fx = 3x - 8$ .

Choice C is incorrect. This is the value of  $f1$ , rather than  $f7$ .

Choice D is incorrect. This is the value of  $f-7$ , rather than  $f7$ .

## Q10

B

B.  $26b + 35g = 881$   
 $b + g = 28$

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Choice B is correct. Outfits for boys cost \$26 each and the teacher ordered  $b$  outfits for boys, so the teacher spent  $26b$  dollars on outfits for boys. Similarly, outfits for girls cost \$35 each and the teacher ordered  $g$  outfits for girls, so the teacher spent  $35g$  dollars on outfits for girls. Since the teacher spent a total of \$881 on outfits for boys and girls, the equation  $26b + 35g = 881$  must be true. And since the teacher ordered a total of 28 outfits, the equation  $b + g = 28$  must also be true.

Choice A is incorrect and may result from switching the constraint on the total number of outfits with the constraint on the cost of the outfits. Choice C is incorrect and may result from switching the constraint on the total number of outfits with the constraint on the cost of the outfits, as well as switching the cost of the outfits for boys with the cost of the outfits for girls. Choice D is incorrect and may result from switching the cost of the outfits for boys with the cost of the outfits for girls.

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### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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